Master Technical Specification: Project 32k-XAI Cognitive Simulation Platform

Document Status: Aggressive / Strict / Final Architectural Blueprint
Project Leads:

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- **Sabihe Khorshidi** (Lead Cognitive Psychologist, Psychometric Mapping, Empirical Validation)

1. Architectural Genesis and Paradigm Shift

The contemporary landscape of Multi-Agent Reinforcement Learning (MARL) is fundamentally flawed for sociological and psychological simulation. Deep Neural Networks (DNNs) act as impenetrable black boxes, representing human behavior as homogeneous, latent-space approximations devoid of empirical psychological grounding.
Project 32k-XAI proposes an aggressive paradigm shift: a **strictly deterministic, highly heterogeneous, Level-3 Mechanistically Interpretable Multi-Agent Platform**. We will instantiate an ecology of 32,000 autonomous agents. Every agent's "brain" is a unified system of differential equations and Markov Decision Processes (MDPs), the parameters of which are strictly bounded by real-world human covariance matrices. There are no black boxes. Every action, betrayal, trust allocation, and system failure can be traced back to a specific cognitive parameter (e.g., temporal discounting, working memory, psychopathy), yielding a 100% transparent "glass-box" simulation.

1.1 The Transdisciplinary Synergy

The architectural viability of this platform relies on the strict division of labor and epistemological synthesis between the two project leads:

- **Sabihe Khorshidi (Cognitive Psychology Lead):** Responsible for the absolute validity of the psychological constructs. Sabihe oversees the translation of raw behavioral data (BART, Stop-Signal, Daw's Two-Step) into computational parameters. She dictates the *theoretical bounds*—mapping empirical DASS-21 scores to algorithmic exploration temperatures, and SD3 Dark Triad traits to social utility functions. Sabihe ensures the agents are mathematically functioning as humans do, not as hyper-rational economic robots.
- **Kian Mansouri Jamshidi (XAI & Computational Lead):** Responsible for the execution, scaling, and algorithmic translation. Kian engineers the vectorized tensor architectures required to run 32,000 parallel Markov models simultaneously at sub-millisecond latencies. Kian is tasked with building the XAI logging protocols, the Numba/JAX optimization frameworks, and the generative Gaussian Copula that scales Sabihe's validated psychometric covariance matrix into 32,000 unique synthetic arrays.

2. The Empirical Substrate: Hierarchical Parameter Extraction

We reject the generation of arbitrary synthetic agents. The genome of the 32k population must be explicitly tethered to the multi-modal data pipeline secured in Phase 1.
Sabihe's Protocol: Before Kian can initialize the population, the raw TSV/CSV files must undergo Hierarchical Bayesian Analysis (HBA) to extract subject-level posterior distributions.

2.1 Dimensional Mappings

1. **Base Personality (OCEAN) & Dark Triad (SD3):** Act as static global priors. For instance, high Conscientiousness mathematically limits the variance in the delay discounting parameter k.
2. **Psychopathology (DASS-21):** Acts as dynamic modulators. Stress directly throttles the working-memory capacity of the agent, suppressing Model-Based (MB) tree-search algorithms and forcing a reversion to Model-Free (MF) habitual responses.
3. **Social Valuation (Trust/Ultimatum Games):** Calibrates the exact distribution of the Fehr-Schmidt inequity aversion coefficients (α, β).
4. **Risk & Inhibitory Control (UCLA ds000030):**
    - *BART:* Extracted parameters govern the agent's coefficient of risk tolerance ρ.
    - *Stop-Signal:* Data is fitted to a **Drift-Diffusion Model (DDM)** to extract drift rate (v), boundary separation (a), and non-decision time (t_0). This computationally defines the agent's impulsivity under time pressure.
5. **Reinforcement Learning Architecture (Daw's Two-Step Task):** Extracts the master control weight ω, defining the precise ratio of goal-directed vs. habitual learning for every individual.

3. The Generative Engine: Multivariate Gaussian Copula

To scale the empirically extracted parameters (from $N \approx 500$ real humans) to $N = 32,000$ synthetic agents without losing the complex non-linear correlations of human psychology, **Kian** will implement a strict Multivariate Gaussian Copula.

3.1 Mathematical Implementation

Let $X = [X_1, X_2, \dots, X_d]$ be the d-dimensional vector of psychological parameters (e.g., $X_1 = \text{Neuroticism}$, $X_2 = \omega$, $X_3 = \text{Delay Discounting } k$).
Since biological distributions are not purely normal (e.g., discount rates follow gamma distributions, Dark Triad scores are right-skewed), we cannot simply use a multivariate normal generator.

1. **Marginal Transformation:** Convert empirical observations to uniform distributions via their empirical cumulative distribution functions (eCDF): $U_i = F_i(X_i)$.

2. **Gaussian Mapping:** Map these to standard normal variables: $Z_i = \Phi^{-1}(U_i)$.
3. **Covariance Extraction:** Compute the Pearson correlation matrix Σ of the latent variables Z.
4. **32k Synthesis:** Kian will draw 32,000 samples from the multivariate normal distribution $N(0, \Sigma)$.
5. **Inversion:** Apply the inverse CDFs to return the values to their true psychometric spaces: $\hat{X}_i = F_i^{-1}(\Phi(\hat{Z}_i))$.

Rigor Output: The result is a $32,000 \times d$ matrix. Agent #14,052 might have clinically high anxiety (DASS), severely impaired inhibitory control (from Stop-Signal DDM a parameter), and high Machiavellianism. These traits are probabilistically linked exactly as they are in the actual clinical population.

4. The Mechanistic Cognitive Architecture (The "Brain")

Deep neural networks are strictly banned from the decision-making loop. Every XAI agent's brain is a continuous loop of transparent mathematical functions mapped directly to the psychometric matrix.

4.1 Module 1: Fehr-Schmidt Social Utility Function

An agent i's perceived value of a systemic state is not merely its own resource accumulation x_i, but is filtered through its social parameters (derived from Ultimatum/Trust data):

$$U_i(X) = x_i - \frac{\alpha_i}{N-1} \sum_{j \neq i} \max(x_j - x_i, 0) - \frac{\beta_i}{N-1} \sum_{j \neq i} \max(x_i - x_j, 0)$$

- α_i (Envy): Bounds the agent's hostility toward richer neighbors (correlated via Copula to Neuroticism/Psychopathy).
- β_i (Guilt/Altruism): Bounds the agent's willingness to share (correlated to Agreeableness).

4.2 Module 2: State-Space Dual-System Learning (Daw Model)

When navigating the environment to maximize U_i, the agent evaluates paths using a hybrid learning algorithm.

Model-Free (Habitual) System: Updates via simple Temporal Difference (TD) error:

$$Q_{MF}(s,a) \leftarrow Q_{MF}(s,a) + \alpha_{MF} [r + \gamma \max_{a'} Q_{MF}(s',a') - Q_{MF}(s,a)]$$

Model-Based (Goal-Directed) System: Computes forward-planning based on transition probability matrices $P(s' \vert{} s,a)$:

$$Q_{MB}(s,a) = \sum_{s'} P(s' \vert{} s,a) \max_{a'} Q_{MB}(s',a')$$

The Arbitration Node (ω): The ultimate action value is the weighted sum:

$$Q_{net}(s,a) = \omega_i Q_{MB}(s,a) + (1 - \omega_i) Q_{MF}(s,a)$$
Crucial Sabihe-Kian Integration: Sabihe's psychological constraints dictate that ω_i is not static. Kian will program ω_i to dynamically decay under environmental stress (simulated DASS-21 activation). A highly stressed agent loses the cognitive capacity for tree-search (Q_{MB}) and reverts to blind habit (Q_{MF}).

4.3 Module 3: Action Translation (Exploration vs. Exploitation)

Values translate into probabilities via a Softmax function:

$$P(a_k) = \frac{\exp(\tau_i \cdot Q_{net}(s,a_k))}{\sum \exp(\tau_i \cdot Q_{net}(s,a_j))}$$
Where τ_i (Inverse Temperature) is derived from the agent's Openness to Experience and baseline exploration parameters.

5. System Execution and The Massively Parallel Sandbox

The computational bottleneck is running 32,000 complex MDP solvers simultaneously.

5.1 Kian's Tensor Vectorization Protocol

Object-Oriented Programming (OOP) with 32k separate class Agent(): instances via standard Python loops will cause massive CPU bottlenecking. Kian will construct the environment using **Data-Oriented Design (DOD)** via JAX or NumPy/Numba.

- All 32,000 Q-tables are flattened into a global tensor $\mathbf{Q}$ of shape (32000, States, Actions).
- All state transitions and reward calculations are executed as parallel matrix multiplications on the GPU.
- *Topology:* Agents will be linked using adjacency matrices to simulate social graphs (Small-World networks), allowing them to play Trust Games and resource trading simultaneously across the network graph.

6. The Pure XAI Protocol: Mechanistic Interpretability

The ultimate deliverable of this platform is the absolute elimination of algorithmic opacity. The system will possess an **Interrogation API**.
When a macro-phenomenon emerges in the simulation (e.g., an economic crash, a breakdown of trust, a segregation of agent sub-populations), the researcher can freeze the simulation at tick t and query any Agent A_i.
The XAI output will mathematically trace the decision:

1. **Log Output:** "Agent 14,052 betrayed Agent 8,991 at $t=450$."
2. **Utility Trace:** "Calculated Q_{net} for [Betray] was 8.4, Q_{net} for [Cooperate] was 4.2."
3. **Algorithmic Breakdown:** "Despite Q_{MB} showing long-term value for cooperation,

the agent's DASS-21 Stress parameter spiked to 94th percentile due to recent resource loss. This forced arbitration weight $\omega_i \to 0.05$, blinding the agent to the future and relying purely on immediate Model-Free (Q_{MF}) greed."

4. **Social Trace:** "Furthermore, baseline Fehr-Schmidt parameter $\alpha_i = 0.82$ (Envy) generated negative utility for cooperating with an agent perceived to have higher resources."

7. Master Execution Roadmap & Role Assignments

Sprint 1: Empirical Data Ingestion & HBA Fitting

- **Sabihe Khorshidi:** Clean and standardize the HCP OCEAN, SD3, and DASS-21 datasets. Identify extreme outliers. Map the conceptual linkages between psychological traits and algorithmic hyper-parameters.
- **Kian Mansouri Jamshidi:** Write parsers for the BIDS/TSV formats downloaded from OpenNeuro (ds000030, ds003745) and OSF. Implement Bayesian fitting scripts (PyMC or Stan) to extract individual $\alpha, \beta, \omega, \tau, \rho, k$ values from the raw human trial logs.

Sprint 2: The Copula Core

- **Sabihe Khorshidi:** Review the aggregate parameter distributions. Validate that the correlations mathematically make psychological sense (e.g., ensuring high psychopathy correlates with high Fehr-Schmidt envy and low Ultimatum offers).
- **Kian Mansouri Jamshidi:** Build the Multivariate Gaussian Copula pipeline. Feed the validated empirical parameters in, and generate the master 32k_Agent_Genome.h5 matrix.

Sprint 3: The Engine Sandbox

- **Kian Mansouri Jamshidi:** Engineer the JAX/Numba vectorized tensor environment. Build the state-space matrices, the reward distribution algorithms, and the network graph adjacency arrays.
- **Sabihe Khorshidi:** Design the "Scenarios" (e.g., Resource Scarcity, Wealth Inequality shocks). Define how environmental state changes map to the agents' DASS-21 stress triggers.

Sprint 4: The XAI Dashboard & Testing

- **Kian Mansouri Jamshidi:** Build the Interrogation API. Ensure that at every tick, matrices of Q, τ, and ω can be logged and visualized in real-time.
- **Both Leads:** Run the first 32,000-agent systemic test. Monitor the emergent dynamics, pause the simulation, and use the XAI framework to causally explain macro-behaviors using the deterministic micro-equations.